

SOC Assisté par Intelligence Artificielle

Détection Wazuh -> Triage IA (Gemma2) -> TheHive -> Cortex -> MISP -> Notification

Omar Babba - PFA EMSI Tanger - RUN_ID PFA-FINAL-20260718-214637

Chaine logique du pipeline

De la detection a la notification - execution reelle, VM VirtualBox

ETAPE 1 - Wazuh (auditd) detecte une activite suspecte sur l'agent surveille

ETAPE 2 - L'alerte reelle declenche le workflow Shuffle via webhook

ETAPE 3 - Gemma2 9B (Ollama, local) effectue le triage : type, MITRE, criticite

ETAPE 4 - TheHive cree automatiquement un cas avec le triage IA

ETAPE 5 - Cortex enrichit l'IOC (ex: AbuseIPDB)

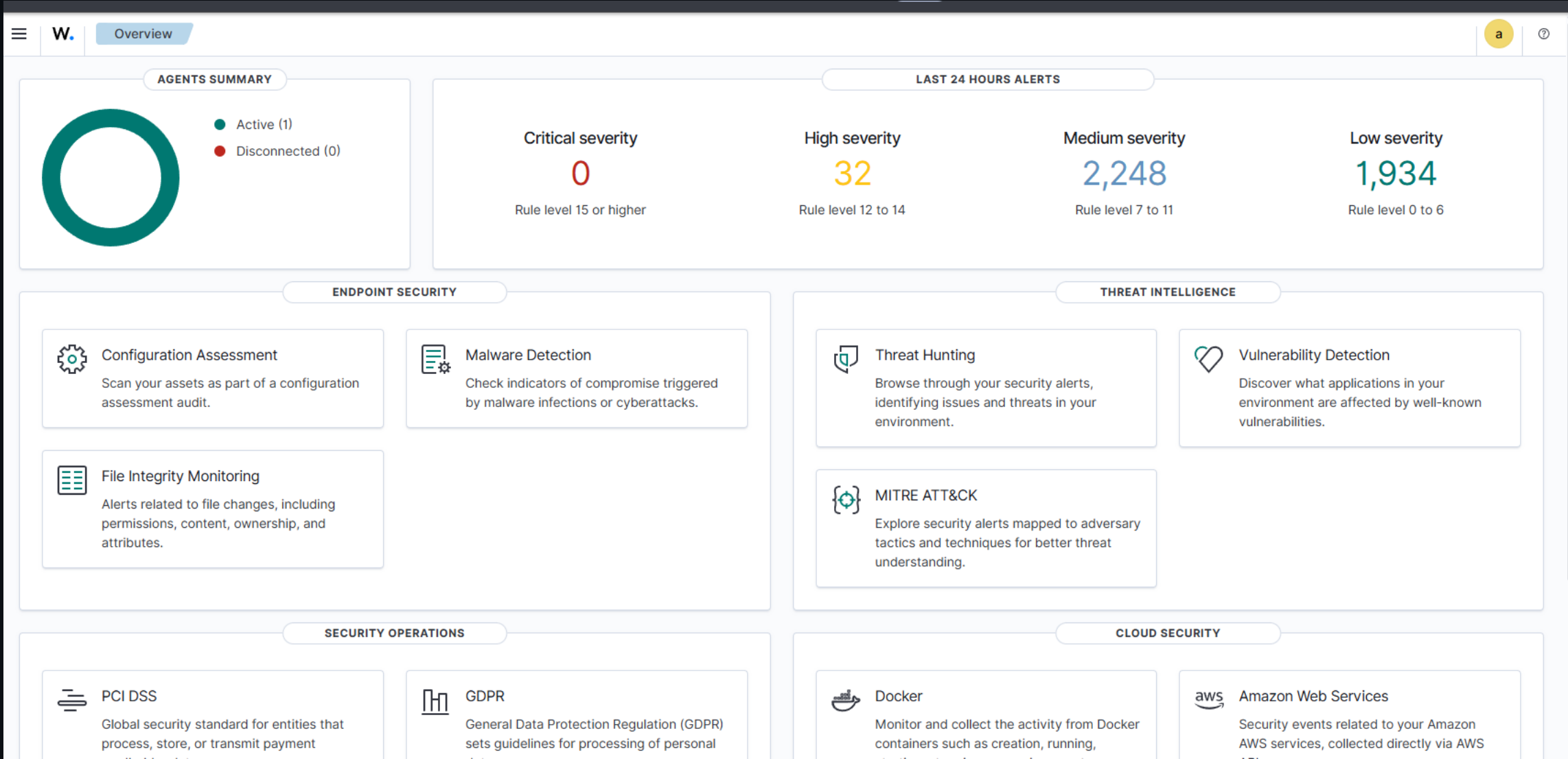
ETAPE 6 - Selon la severite : evenement MISP (haute) ou tag simple (basse)

ETAPE 7 - Notification finale envoyee

Chaque etape critique est protegee par une garde de statut HTTP.

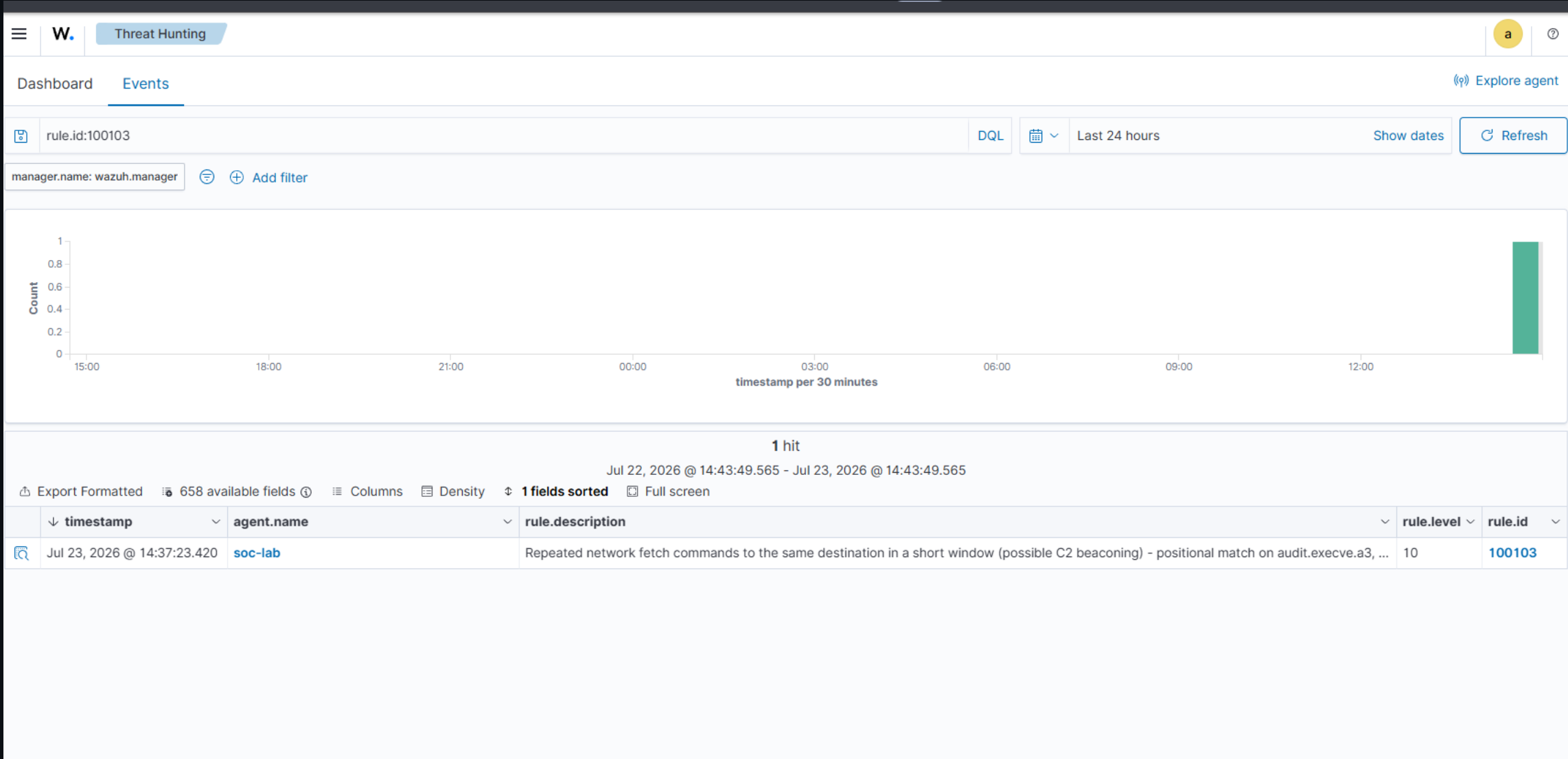
ETAPE 1 / 7 - Wazuh detecte

Dashboard reel : agent actif, alertes classees par severite



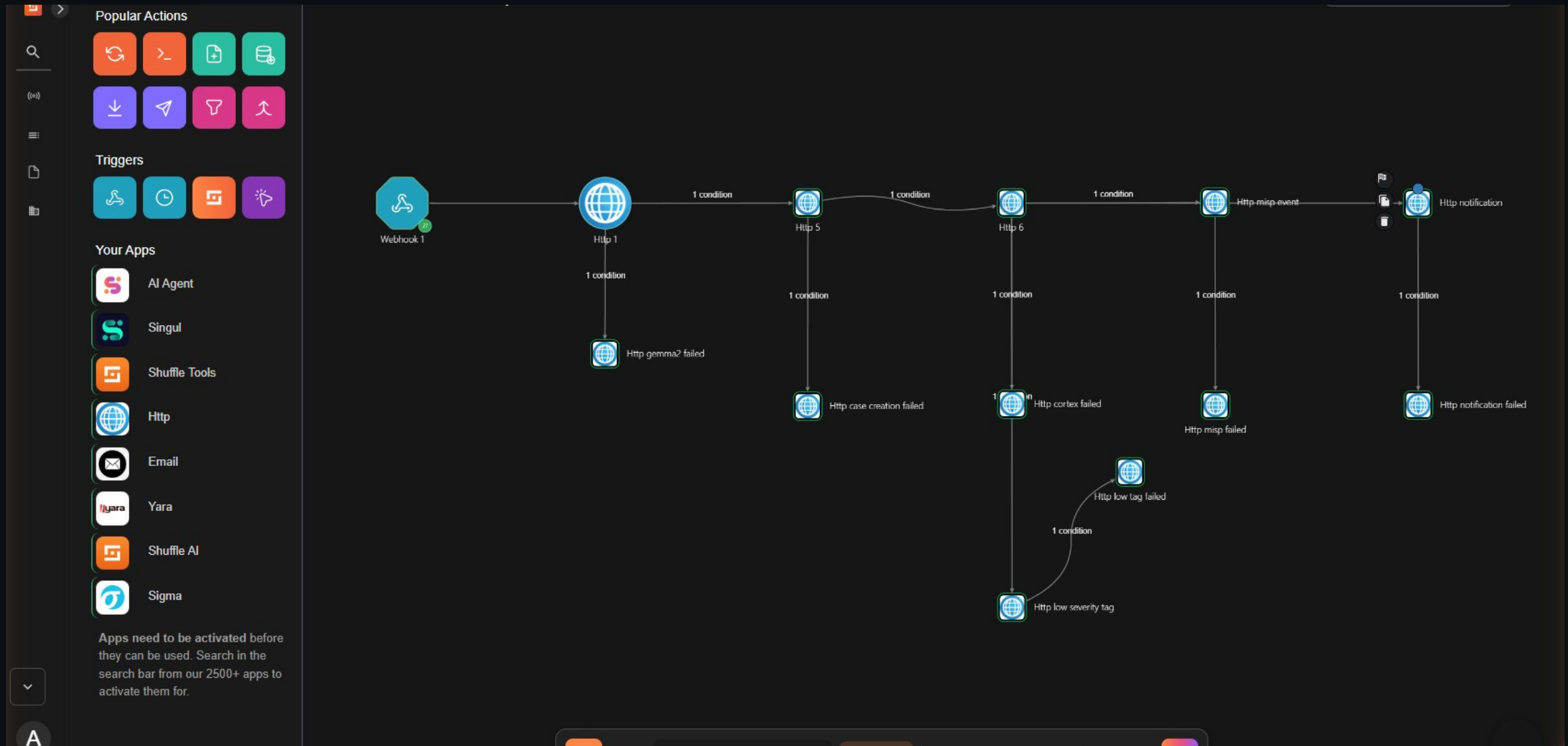
ETAPE 1 / 7 - Alerte reelle declenchee

Regle 100103 - C2 beaconing - MITRE T1071 - agent soc-lab



ETAPE 2-3 / 7 - Shuffle recoit l'alerte et orchestre

Webhook -> Gemma2 (triage) -> TheHive -> Cortex -> MISP/tag -> notification



ETAPE 3 / 7 - Triage par IA (Gemma2 9B)

Prompt reel envoye au modele local Ollama pour classification MITRE

Search apps, triggers...

Popular Actions

Triggers

Your Apps

Apps need to be activated before they can be used. Search in the search bar from our 2500+ apps to activate them for.

PFA-FINAL-20260718-214637 -
Orchestration complete SOC-IA

Build Debug

```
graph LR; W[Webhook 1] --> H1[Http 1]; H1 -- "1 condition" --> H5[Http 5]; H1 -- "1 condition" --> H1_err[Http gemma2 failed]; H5 -- "1 condition" --> H6[Http 6]; H5 -- "1 condition" --> H5_err[Http case creation failed]; H6 -- "1 condition" --> H7[Http 7]; H6 -- "1 condition" --> H6_err[Http cortex failed]; H7 -- "1 condition" --> H7_err[Http misp fail]; H7 -- "1 condition" --> H7_low[Http low tag failed]; H7 -- "1 condition" --> H7_low_sev[Http low severity tag];
```

Setup Configuration

POST

Url *

http://172.21.0.1:11434/api/generate

Optional Parameters

Body

```
{ "model": "gemma2:9b-instruct-q4_0", "prompt": "Analyse cette alerte de securite Wazuh et reponds uniquement en JSON avec les champs incident_type, criticite, mitre_tactic, mitre_technique," }
```

Headers

Content-Type: application/json

Test Action

Username

ETAPE 4 / 7 - TheHive cree le cas

Triage brut de Gemma2 visible dans la description du cas reel

→

#22 [SHUFFLE-AUTO] Triage automatisé

📁

🔍

📋

🔔

📌

🔍

🏠

🧠

💬

🔧 id ~28720

👤 Created by SOC Pipeline 5.2 Service Account

📅 Created at 23/07/2026 00:47

SEVERITY:MEDIUM

TLP:AMBER

PAP:AMBER

Assignee [Assign to me](#)

S

SOC Pipeline 5.2 Service Account

Status

New

Start date

23/07/2026 00:47

Tasks completion

No tasks

Contributors

S

General

Tasks (0)

Observables (0)

TTPs (0)

Attachments

Timeline

Pages

History

Linked alerts

X

* Title

[SHUFFLE-AUTO] Triage automatisé

Tags

shuffle-auto

PFA-FINAL-20260718-214637

Description

Cas cree automatiquement par le workflow Shuffle Orchestration complete SOC-IA.
Sortie Gemma2 (brute): {"incident_type": "Privilege Escalation", "criticite": "High", "mitre_tactic": "Defense Evasion", "mitre_technique": "T1059", "resume": "Modification du fichier sudoers par un compte non autorisé, ajoutant l'utilisateur 'backdoor' au groupe sudo. ", "recommandation": "Investigate the user 'backdoor' and its activities. Review the sudoers file for unauthorized changes. Implement stricter access controls and multi-factor authentication."}

Comments

🖨️

Preuve : le cas TheHive vient bien de Shuffle

Reponse HTTP reelle 201 de l'execution Shuffle - meme case id ~28720 / #22

```
PEA FINAL 20260718 214637

Execution reelle 48297155-9f5c-412e-90d8-3c9b08f3773b – noeud http_5 (creation cas TheHive)

Status noeud Shuffle: SUCCESS | Demarree: 23/07/2026 00:46:05 | Terminee: 00:47:30

{
  "status": 201,
  "body": {
    "_id": "~28720",
    "_type": "Case",
    "_createdBy": "soc-pipeline52@thehive.local",
    "_createdAt": 1784764044367,
    "number": 22,
    "title": "[SHUFFLE-AUTO] Triage automatise",
    "description": "Cas cree automatiquement par le workflow Shuffle Orchestration complete SOC-IA.\n\nSortie Gemma2 (brute): {\n\"incident_type\": \"Privilege Escalation\", \"criticite\": \"High\", \"mitre_tactic\": \"Defense Evasion\", \"mitre_technique\": \"T1059\", \"resume\": \"Modification du fichier sudoers par un compte non autoris\u00e9, ajoutant l'utilisateur 'backdoor' au groupe sudo. \", \"recommandation\": \"Investigate the user 'backdoor' and its activities. Review the sudoers file for unauthorized changes. Implement stricter access controls and multi-factor authentication.\"},\n\"severity\": 2,\n\"severityLabel\": \"MEDIUM\", \"startDate\": 1784764044258,\n\"tags\": [\n  \"shuffle-auto\", \"PFA-FINAL-20260718-214637\"\n],\n\"flag\": false,\n\"tlp\": 2,\n\"tlpLabel\": \"AMBER\", \"pap\": 2,\n\"papLabel\": \"AMBER\", \"status\": \"New\", \"stage\": \"New\", \"assignee\": \"soc-pipeline52@thehive.local\", \"customFields\": [],\n\"userPermissions\": [\n  \"manageCase/create\", \"manageAlert/update\", \"manageProcedure\", \"managePage\", \"manageObservable\",
```

Correle avec la capture precedente : createdBy soc-pipeline52@thehive.local (compte de service), pas un humain.

ETAPE 5 / 7 - Cortex enrichit l'IOC

Historique des analyses AbuseIPDB reelles, toutes en Success

 [+ New Analysis](#)

[Jobs History](#) [Analyzers](#) [Responders](#) [Organization](#)

SLO

 soc-lab/SOC Lab OrgAdmin

Jobs History (12)

Data Types (6)
Select ▾

Job Type (2)
Select ▾

Analyzers (3)
Select ▾

Observable
Search for observable data

Search

Clear

Pagination
50 / page ▾

Status	Job details	TLP	PAP
Success	[ip] 185[.]220[.]101[.]7 Analyzer: AbuseIPDB_2_0 Date: 9 minutes ago User: soc-lab/orgadmin	TLP:AMBER	PAP:AMBER
	View Delete		
Success	[ip] 185[.]220[.]101[.]7 Analyzer: AbuseIPDB_2_0 Date: 23 minutes ago User: soc-lab/orgadmin	TLP:AMBER	PAP:AMBER
	View Delete		
Success	[ip] 185[.]220[.]101[.]7 Analyzer: AbuseIPDB_2_0 Date: 14 hours ago User: soc-lab/orgadmin	TLP:AMBER	PAP:AMBER
	View Delete		
Success	[ip] 185[.]220[.]101[.]7 Analyzer: AbuseIPDB_2_0 Date: 14 hours ago User: soc-lab/orgadmin	TLP:AMBER	PAP:AMBER
	View Delete		
Success	[ip] 185[.]220[.]101[.]7 Analyzer: AbuseIPDB_2_0 Date: 14 hours ago User: soc-lab/orgadmin	TLP:AMBER	PAP:AMBER
	View Delete		
Success	[ip] 185[.]220[.]101[.]7 Analyzer: AbuseIPDB_2_0 Date: 14 hours ago User: soc-lab/orgadmin	TLP:AMBER	PAP:AMBER
	View Delete		

ETAPE 5 / 7 - Rapport d'enrichissement detaille

Resultat reel de l'API AbuseIPDB sur l'IP suspecte

Cortex

+ New Analysis

Jobs History

Analyzers

Responders

Organization

SLO soc-lab/SOC Lab OrgAdmin

Job details

Back to list

AbuseIPDB_2_0

Artifact

[IP] 185[.]220[.]101[.]7

Date

9 minutes ago

TLP

TLP:AMBER

PAP

PAP:AMBER

Status

Success

Report summary

AbuseIPDB:Tor=True AbuseIPDB:Usage=Fixed Line ISP AbuseIPDB:Score=1

Job report

Parameters

{}

Report

```
{  "summary": {    "taxonomies": [      {        "level": "info",        "namespace": "AbuseIPDB",        "predicate": "Tor",        "value": "True"      },      {        "level": "info",        "namespace": "AbuseIPDB",        "predicate": "Usage",        "value": "Fixed Line ISP"      },      {        "level": "malicious",        "namespace": "AbuseIPDB",        "predicate": "Score",        "value": 100      },      {        "level": "malicious",        "namespace": "AbuseIPDB",        "predicate": "Reports",        "value": 67      }    ]  }
```

ETAPE 6 / 7 - MISP partage la menace

Evenement cree automatiquement par le pipeline (severite haute)

Home

Event Actions

Dashboard

Galaxies

Input Filters

Global Actions

Sync Actions

Administration

Logs

API

Bookmarks

★

MISP

Admin

✉

Log out

View Event

View Correlation Graph

View Event History

Edit Event

Delete Event

Add Attribute

Add Object

Add Attachment

Add Event Report

Populate from...

Enrich Event

Merge attributes from...

Publish Event

Publish (no email)

Delegate Publishing

Run Ad-Hoc Workflow

Recorrelate Event

Contact Reporter

Download as...

Add Event to Collection

List Events

Add Event

[SHUFFLE-AUTO] Triage automatisé - PFA-FINAL-20260718-214637

Event ID

11

UUID

479f1e5c-24d7-451a-977c-256069308e1d

Creator org

SOC-LAB-PFA

Owner org

SOC-LAB-PFA

Creator user

admin@soc-lab.local

Protected Event (experimental)

Event is in unprotected mode.

Switch to protected mode

Tags

Date

2026-07-22

Threat Level

Medium

Analysis

Initial

Distribution

Your organisation only

Warnings

Contextualisation:

Your event has neither tags nor galaxy clusters attached - generally adding context to an event allows for quicker decision making and more accurate filtering, it is highly recommended that you label your events to the best of your ability.

Published

No

#Attributes

1 (0 Objects)

First recorded change

2026-07-22 23:47:29

Last change

2026-07-22 23:47:29

Modification map

Sightings

0 (0) - restricted to own organisation only.

— Pivots

— Galaxy

+ Event graph

+ Event timeline

+ Correlation graph

+ Galaxy matrix

+ Event reports

— Attributes

— Discussion

Related Events

Order by date

SOC-LAB-PFA

[SHUFFLE-AUTO] Triage automatisé - PFA-FINAL-20260718-214637

2026-07-23

1

SOC-LAB-PFA

[SHUFFLE-AUTO] Triage automatisé - PFA-FINAL-20260718-214637

2026-07-22

1

SOC-LAB-PFA

[SHUFFLE-AUTO] Triage automatisé - PFA-FINAL-20260718-214637

2026-07-22

1

SOC-LAB-PFA

[SHUFFLE-AUTO] Triage automatisé - PFA-FINAL-20260718-214637

2026-07-22

1

SOC-LAB-PFA

[SHUFFLE-AUTO] Triage automatisé - PFA-FINAL-20260718-214637

2026-07-22

1

SOC-LAB-PFA

[SHUFFLE-AUTO] Triage automatisé - PFA-FINAL-20260718-214637

2026-07-22

1

ETAPE 6 / 7 - IOC reel extrait et correle

Attribut ip-dst correle a 4 autres evenements, IDS actif

Home

Event Actions

Dashboard

Galaxies

Input Filters

Global Actions

Sync Actions

Administration

Logs

API

Bookmarks

★

MISP

Admin

Log out

Contact Reporter

Download as...

Add Event to Collection

List Events

Add Event

Published

No

#Attributes

1 (0 Objects)

First recorded change

2026-07-22 23:47:29

Last change

2026-07-22 23:47:29

Modification map

Sightings

0 (0) - restricted to own organisation only.

—Pivots

—Galaxy

+Event graph

+Event timeline

+Correlation graph

+Galaxy matrix

+Event reports

—Attributes

—Discussion

×

11: [SHUFFLE-AUT...

Galaxies

+

+

« previous

next »

view all

+

☰

☰

☰

☰

Scope toggle

Deleted

Decay score

Context

Related Tags

Filtering tool

Expand all Objects

Collapse all Attributes

Enter value to search

Q

×

☐	Date ↑	Context	Category	Type	Value	Tags	Galaxies	Comment	Correlate	Related Events	Feed hits	IDS	Distribution	Sightings	Activity	Actions
☐	2026-07-22*	590...d68	Network activity	ip-dst	185.220.101.7	+ +	+ +	IOC extrait automatiquement du triage Gemma2 (workflow Shuffle)	☒	7 8 9 10 Show 2 more...	☒	☒	Inherit	👍 🗨 ✎	🗨 🗑 📄 🗑	

Page 1 of 1, showing 1 records out of 1 total, starting on record 1, ending on 1

« previous

next »

view all

Discussion

Quote

Event

Thread

Link

Code

ETAPE 7 / 7 - Notification finale

Dernier noeud du pipeline : alerte envoyee au canal de notification

Search apps, triggers...

Popular Actions

Triggers

Your Apps

Apps need to be activated before they can be used. Search in the search bar from our 2500+ apps to activate them for.

PFA-FINAL-20260718-214637 -
Orchestration complete SOC-IA

Build Debug

```
graph LR; W1[Webhook 1] -- "1 condition" --> H1[Http 1]; H1 -- "1 condition" --> H5a[Http 5]; H5a -- "1 condition" --> H5b[Http 5]; H5b -- "1 condition" --> HME[Http miss event]; HME -- "1 condition" --> HN[Http notification]; H1 -- "1 condition" --> G1(( )); G1 -- "1 condition" --> G2(( )); G2 -- "1 condition" --> G3(( )); G3 -- "1 condition" --> G4(( )); G4 -- "1 condition" --> G5(( )); G5 -- "1 condition" --> G6(( )); G6 -- "1 condition" --> G7(( )); G7 -- "1 condition" --> G8(( )); G8 -- "1 condition" --> G9(( )); G9 -- "1 condition" --> G10(( )); G10 -- "1 condition" --> G11(( )); G11 -- "1 condition" --> G12(( )); G12 -- "1 condition" --> G13(( )); G13 -- "1 condition" --> G14(( )); G14 -- "1 condition" --> G15(( )); G15 -- "1 condition" --> G16(( )); G16 -- "1 condition" --> G17(( )); G17 -- "1 condition" --> G18(( )); G18 -- "1 condition" --> G19(( )); G19 -- "1 condition" --> G20(( )); G20 -- "1 condition" --> G21(( )); G21 -- "1 condition" --> G22(( )); G22 -- "1 condition" --> G23(( )); G23 -- "1 condition" --> G24(( )); G24 -- "1 condition" --> G25(( )); G25 -- "1 condition" --> G26(( )); G26 -- "1 condition" --> G27(( )); G27 -- "1 condition" --> G28(( )); G28 -- "1 condition" --> G29(( )); G29 -- "1 condition" --> G30(( )); G30 -- "1 condition" --> G31(( )); G31 -- "1 condition" --> G32(( )); G32 -- "1 condition" --> G33(( )); G33 -- "1 condition" --> G34(( )); G34 -- "1 condition" --> G35(( )); G35 -- "1 condition" --> G36(( )); G36 -- "1 condition" --> G37(( )); G37 -- "1 condition" --> G38(( )); G38 -- "1 condition" --> G39(( )); G39 -- "1 condition" --> G40(( )); G40 -- "1 condition" --> G41(( )); G41 -- "1 condition" --> G42(( )); G42 -- "1 condition" --> G43(( )); G43 -- "1 condition" --> G44(( )); G44 -- "1 condition" --> G45(( )); G45 -- "1 condition" --> G46(( )); G46 -- "1 condition" --> G47(( )); G47 -- "1 condition" --> G48(( )); G48 -- "1 condition" --> G49(( )); G49 -- "1 condition" --> G50(( )); G50 -- "1 condition" --> G51(( )); G51 -- "1 condition" --> G52(( )); G52 -- "1 condition" --> G53(( )); G53 -- "1 condition" --> G54(( )); G54 -- "1 condition" --> G55(( )); G55 -- "1 condition" --> G56(( )); G56 -- "1 condition" --> G57(( )); G57 -- "1 condition" --> G58(( )); G58 -- "1 condition" --> G59(( )); G59 -- "1 condition" --> G60(( )); G60 -- "1 condition" --> G61(( )); G61 -- "1 condition" --> G62(( )); G62 -- "1 condition" --> G63(( )); G63 -- "1 condition" --> G64(( )); G64 -- "1 condition" --> G65(( )); G65 -- "1 condition" --> G66(( )); G66 -- "1 condition" --> G67(( )); G67 -- "1 condition" --> G68(( )); G68 -- "1 condition" --> G69(( )); G69 -- "1 condition" --> G70(( )); G70 -- "1 condition" --> G71(( )); G71 -- "1 condition" --> G72(( )); G72 -- "1 condition" --> G73(( )); G73 -- "1 condition" --> G74(( )); G74 -- "1 condition" --> G75(( )); G75 -- "1 condition" --> G76(( )); G76 -- "1 condition" --> G77(( )); G77 -- "1 condition" --> G78(( )); G78 -- "1 condition" --> G79(( )); G79 -- "1 condition" --> G80(( )); G80 -- "1 condition" --> G81(( )); G81 -- "1 condition" --> G82(( )); G82 -- "1 condition" --> G83(( )); G83 -- "1 condition" --> G84(( )); G84 -- "1 condition" --> G85(( )); G85 -- "1 condition" --> G86(( )); G86 -- "1 condition" --> G87(( )); G87 -- "1 condition" --> G88(( )); G88 -- "1 condition" --> G89(( )); G89 -- "1 condition" --> G90(( )); G90 -- "1 condition" --> G91(( )); G91 -- "1 condition" --> G92(( )); G92 -- "1 condition" --> G93(( )); G93 -- "1 condition" --> G94(( )); G94 -- "1 condition" --> G95(( )); G95 -- "1 condition" --> G96(( )); G96 -- "1 condition" --> G97(( )); G97 -- "1 condition" --> G98(( )); G98 -- "1 condition" --> G99(( )); G99 -- "1 condition" --> G100(( ));
```

Setup Configuration

POST

Url *

http://172.21.0.1:9090/notify

Optional Parameters

Body

```
{ "channel": "soc-alerts (simulation locale)", "message": "[NOTIFICATION AUTO] Workflow Shuffle Orchestration complete SOC-IA - RUN_ID PFA-FINAL-20260718-214637. Triage Gemma2" }
```

Headers

Content-Type: application/json

Test Action

Username

Bilan

Tests reels effectues le jour meme

3 executions completes reelles du pipeline aujourd'hui

Run 100% vert : Gemma2 -> TheHive -> Cortex -> MISP -> notification

Gardes d'echec verifiees en conditions reelles (TheHive 401, Cortex ES down)

Aucune donnee simulee : alertes, cas, jobs et evenements sont reels